

Hospital Patient Data Analysis Using SQL

Portfolio Project Report

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1. Introduction

Healthcare organizations generate large volumes of operational data every day. This project demonstrates how SQL can be used to transform raw patient records into meaningful business insights. Using a dataset of 200 patient records, the analysis explores patient demographics, billing patterns, diagnosis trends, ward utilization, and revenue distribution.

2. Project Objectives

- Determine the overall size and characteristics of the patient population.
- Analyze demographic patterns such as age and gender.
- Identify cities contributing the highest number of patients and revenue.
- Compare diagnoses and wards using aggregate analysis.
- Segment patients based on billing levels.
- Identify high-cost patients using subqueries.
- Generate business insights to support hospital management.

3. Tools & Skills Used

Tools Used

- MySQL Workbench
- SQL
- Microsoft Excel (Dataset Preparation)
- GitHub

SQL Concepts Used

- SELECT
- WHERE
- Aggregate Functions
- GROUP BY
- HAVING
- ORDER BY
- CASE Statements
- Subqueries
- Common Table Expressions (CTEs)

4. Dataset Description

Attribute	Meaning
PatientID	Unique patient identifier
Name	Patient name
Age	Age in years
Gender	Male/Female
City	Patient residence
Diagnosis	Medical condition
Ward	Hospital department
BillAmount	Treatment cost

5. Dataset Summary

Dataset Information	Value
Total Records	200
Total Columns	8
Cities	5
Diagnoses	8
Wards	5
Gender Categories	2

6. SQL Analysis and Business Interpretation

Q1. How many Patients are in Hospital?

SQL Query:

```
SELECT COUNT(*) AS TotalPateints
```

```
FROM Patient;
```

Result:

200

Interpretation:

The database contains 200 patient records, indicating a sufficiently large dataset for demonstrating SQL analysis techniques.

Q2 What is the average age of patients?

SQL Query:

```
SELECT AVG(Age) AS Avg_Age  
  
FROM Patient;Average Age
```

Result:

48.4550 years

Interpretation:

The average patient age is approximately 48 years, suggesting that middle-aged and older adults form a significant portion of the hospital population.

Q3. How many patients belongs to each gender?

SQL Query:

```
SELECT Gender,  
  
        COUNT(*) AS TotalPatients  
  
FROM Patient  
  
GROUP BY Gender;
```

Result:

Male: 104 | Female: 96

Interpretation:

The gender distribution is relatively balanced with a slight majority of male patients.

Q4. Which city as the most patients?

SQL Query:

```
SELECT City,  
  
        COUNT(*) AS TotalPatients  
  
FROM Patient  
  
GROUP BY City  
  
ORDER BY TotalPatients DESC;
```

Result:

Islamabad 46, Peshawar 44, Haripur 42, Abbottabad 40, Rawalpindi 28

Interpretation:

Islamabad has the highest patient count, making it the hospital's largest represented service area.

Q5. Which diagnosis has the highest average bill?**SQL Query:**

```
SELECT Diagnosis,  
  
        AVG(Bill) AS AvgBill  
  
FROM Patient  
  
GROUP BY Diagnosis  
  
ORDER BY AvgBill DESC;
```

Result:

Infection has the highest average bill (62,201.28)

Interpretation:

Patients diagnosed with infections incur the highest average treatment cost, which may reflect longer stays or more resource-intensive care.

Q6. Which ward have more than 20 patients?**SQL Query:**

```
SELECT Ward,  
  
        COUNT(*) AS TotalPatients  
  
FROM Patient  
  
GROUP BY Ward  
  
HAVING COUNT(*) > 20;
```

Result:

Ward	Number of Patients
General	52
Pediatrics	43
Cardiology	37
Orthopedics	34

Ward	Number of Patients
ICU	34

Interpretation

The analysis shows that **all five wards have more than 20 patients**, satisfying the query condition. Among them, the **General Ward** has the highest patient count with **52 patients**, making it the busiest ward in the hospital. It is followed by **Pediatrics (43 patients)**, **Cardiology (37 patients)**, while both **Orthopedics** and **ICU** have **34 patients** each. These findings suggest that the General Ward requires the greatest allocation of staff and resources to effectively manage patient demand, while the other wards also maintain substantial patient volumes and should be adequately supported.

Q7. Categorize patients based on their bill?

SQL Query:

```
SELECT Name,
        Bill,
        CASE
        WHEN Bill < 30000 THEN 'LOW'
        WHEN Bill BETWEEN 30000 AND 70000 THEN 'MEDIUM'
        ELSE 'HIGH'
        END AS Bill_Category
FROM Patient;
```

Result:

Patient Name	Bill Amount	Category
Sara Malik	74,898	HIGH
Maryam Siddiqui	119,889	HIGH
Ahmed Ahmed	22,974	LOW
Komal Raza	14,508	LOW
Ayesha Sheikh	92,749	HIGH
Sana Khan	38,175	MEDIUM

Patient Name	Bill Amount	Category
Waqas Raza	57,200	MEDIUM
Bilal Khan	97,579	HIGH
Fatima Butt	51,742	MEDIUM
Saad Ahmed	77,620	HIGH

(The same classification was applied to all 200 patient records.)

Interpretation:

The **CASE** statement effectively segmented patients into three billing groups based on their treatment costs. The results show that the dataset contains a mix of **Low**, **Medium**, and **High** bill amounts, indicating variation in the complexity and cost of medical services provided. Patients in the **High** category represent cases with more expensive treatments or longer care requirements, while those in the **Low** category generally incurred relatively modest medical expenses. This type of classification can support budgeting, financial reporting, and the identification of high-cost cases for further review.

Q8. Find patients whose bill is above the overall average?

SQL Query:

```
SELECT Name,
        Bill
FROM Patient
WHERE Bill > (
                SELECT AVG(Bill)
                FROM Patient);
```

Result:

The query identified **90 patients** whose treatment bill is **higher than the overall average bill amount** in the dataset. Some of the patients returned by the query include:

Patient Name	Bill Amount
Sara Malik	74,898
Maryam Siddiqui	119,889

Patient Name	Bill Amount
Ayesha Sheikh	92,749
Waqas Raza	57,200
Bilal Khan	97,579
Saad Ahmed	77,620
Zainab Ahmed	72,821
Yasir Butt	108,332
Sana Ahmed	116,357
Zainab Siddiqui	72,473

Total patients returned by the query: 90

Interpretation:

This analysis uses a **subquery** to compare each patient's bill with the **overall average bill amount** in the hospital dataset. The results show that **90 out of 200 patients (45%)** have treatment costs that exceed the average. This indicates that a substantial proportion of patients required relatively expensive medical services. Identifying these higher-cost cases can help hospital administrators monitor resource utilization, evaluate treatment expenses, and develop strategies for financial planning and cost management. It also enables management to investigate whether certain diagnoses, wards, or patient groups are consistently associated with above-average healthcare costs.

Q9. Create a temporary list of high-bill patient?

SQL Query:

WITH HighBillsPatients AS

(

SELECT *

FROM Patient

WHERE Bill > 80000

)

SELECT *

FROM HighBillsPatients;

Result:

The query returned **53 patients** whose **Bill Amount exceeds 80,000**. The result includes patient details such as Patient ID, Name, Age, Gender, City, Diagnosis, Ward, and Bill Amount.

A few examples from the output are shown below:

Patient ID	Patient Name	Diagnosis	Ward	Bill Amount
2	Maryam Siddiqui	Asthma	Orthopedics	119,889
14	Sana Ahmed	Hypertension	Cardiology	116,357
18	Sana Butt	Diabetes	Orthopedics	117,227
46	Noor Qureshi	Allergy	Cardiology	115,825
60	Komal Ahmed	Fracture	Pediatrics	114,983
114	Sana Malik	Asthma	Pediatrics	117,884
138	Hina Sheikh	Infection	General	118,278
148	Daniyal Sheikh	Infection	Orthopedics	115,932
168	Haris Ahmed	Flu	Pediatrics	119,419
193	Yasir Khan	Allergy	Cardiology	115,062

Interpretation:

This analysis identifies patients whose treatment costs are **greater than 80,000**, highlighting cases that required relatively expensive medical care. Out of the 200 patients in the dataset, **53 patients (26.5%)** fall into this high-billing category. These patients are distributed across different cities, diagnoses, and hospital wards, indicating that high treatment costs are not limited to a single medical condition or location. Such information can help hospital management monitor costly cases, improve financial planning, allocate resources efficiently, and investigate factors contributing to higher healthcare expenses.

Q10. Which city generates the highest hospital revenue?

SQL Query:

```
SELECT City,
        SUM(Bill) AS TotalRevenue
FROM Patient
GROUP BY City
```


ORDER BY TotalRevenue DESC;

Result:

Islamabad	2549387
Peshawar	2491453
Abbottabad	2293554
Haripur	2123750
Rawalpindi	1484378

Interpretation:

Islamabad contributes the highest overall revenue, highlighting its importance from both operational and financial perspectives.

Q11. What are the top 10 highest bill patients?

SQL Query:

SELECT Patient_ID,

Name,

Diagnosis,

Bill

FROM Patient

ORDER BY Bill DESC

LIMIT 10;

Result:

2	Maryam Siddiqui	Asthma	119889
168	Haris Ahmed	Flu	119419
128	Haris Hussain	Fracture	118525
138	Hina Sheikh	Infection	118278
114	Sana Malik	Asthma	117884
18	Sana Butt	Diabetes	117227
14	Sana Ahmed	Hypertension	116357
148	Daniyal Sheikh	Infection	115932
46	Noor Qureshi	Allergy	115825
193	Yasir Khan	Allergy	115062

Interpretation:

A small subset of patients accounts for substantial treatment expenditure, which is useful for revenue analysis.

Q12. Which age group generates the most revenue?

SQL Query:

SELECT

CASE

WHEN Age < 30 THEN '18 - 29'

WHEN Age BETWEEN 30 AND 50 THEN '30 - 50'

ELSE '51+'

END AS AgeGroup,

SUM(Bill) AS TotalRevenue

FROM Patient

GROUP BY AgeGroup

ORDER BY TotalRevenue DESC;

Result:

51+	5820296
30 - 50	3297828
18 - 29	1824398

Interpretation:

Patients aged 51 years and above contribute the largest share of revenue, reflecting increased healthcare utilization among older adults.

5. Overall Findings

- Total records analyzed: 200.
- Average patient age: 48.46 years.
- Male patients slightly outnumber female patients.
- Islamabad leads in both patient volume and total revenue.
- Infection has the highest average treatment bill.
- The General Ward handles the highest patient load.
- Patients aged 51+ contribute the highest revenue.

6. Conclusion

This project demonstrates the practical application of SQL for healthcare data analysis using a realistic hospital dataset containing 200 patient records. SQL was used to analyze patient demographics, treatment costs, hospital revenue, ward utilization, and diagnosis trends through a series of business-oriented questions. The project applied key SQL concepts, including aggregate functions, GROUP BY, HAVING, CASE statements, subqueries, Common Table Expressions (CTEs), and sorting techniques to extract meaningful insights from the data. The analysis revealed that Islamabad generated the highest hospital revenue, the General Ward served the largest number of patients, Infection had the highest average treatment cost, and patients aged 51 years and above contributed the greatest share of total revenue. These findings demonstrate how SQL can transform raw healthcare data into valuable information that supports operational decision-making, resource allocation, financial planning, and performance evaluation.